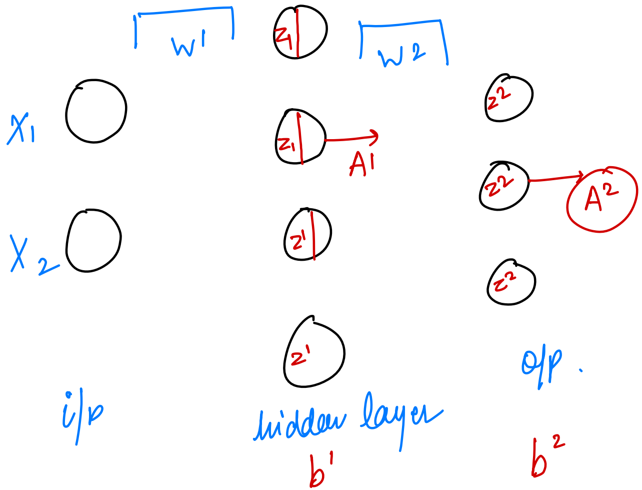


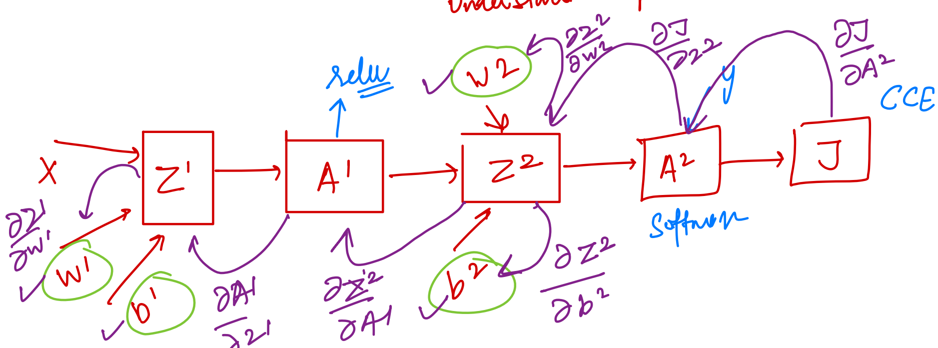
$A^1 = \text{Act}(z^1)$

non-linear

understand complex relatn

Relu → hidden

Softmax → output layer



$$\frac{\partial J}{\partial W^1} = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial A^1} \times \frac{\partial A^1}{\partial Z^1} \times \frac{\partial Z^1}{\partial W^1}$$

$A^2 - y$; ΔZ^2 ; W^2 ; ΔA^1 ; ΔZ^1 ; relu derivative

$$\frac{\partial J}{\partial b^1} = \Delta Z^1 \times \frac{\partial Z^1}{\partial b^1}$$

$$\frac{\partial J}{\partial W^2} = \Delta W_2 = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial W^2} = (A^2 - y) \cdot A^1 = \Delta W^2$$

pred output ; i/p from prev layer

$$\Delta Z^2 \cdot A^1$$

$$\frac{\partial J}{\partial b^2} = \Delta b_2 = \frac{\partial J}{\partial A^2} \times \frac{\partial A^2}{\partial Z^2} \times \frac{\partial Z^2}{\partial b^2} = (A^2 - y)$$

✓ $Z^2 = W^2(A^1) + b^2$

$$\frac{\partial Z^2}{\partial W^2} = A^1$$

$$\frac{\partial Z^2}{\partial A^1} = W^2$$

$A^1 = \text{Relu}(Z^1)$; $\max(0, Z)$

$$\frac{\partial A^1}{\partial Z^1} = \begin{cases} Z & Z > 0 \\ 0 & Z \le 0 \end{cases}$$

$Z > 0 \rightarrow 1$

$Z \le 0 \rightarrow 0$

$\boxed{Z > 0}$

$Z \le 0$

$$\frac{\partial J}{\partial W^1} = (A^2 - y) \cdot W^2 \cdot 1 \cdot X$$

0

$$\boxed{\Delta W^1 = \Delta Z^1 \cdot X}$$

$$\Delta b^1 = \Delta Z^1$$

$$\frac{\partial Z^1}{\partial W^1} = W^1 X + b$$

$$= X$$